Ex 14) Packet sniffing using Raw Sockets.

Aim:

To implement a paket sniffing using python raw sockets and display the Source IP, destination IP and protocal of captured packets.

Code:

```
import socket
import struct
sock = socket.socket( socket.AF_INET,
                      socket.SOCK_RAW,
                      socket.IPPROTO_IP)

while TRUE:
    packet, address = sock.recvfrom(65535)
    ip_header = packet[0:20]
    iph = struct.unpack("!BBHHHBBH4s4s", ip_header)

    protocol = iph[6]
    print("Source IP:", source_ip)
    Print("Destination Ip:", destination_ip)
    Print("protocol:", protocol)
```

Sample output:

```
Packet Sniffer Started...
Waited for packets ...
-----------------------
Source Ip : 192.168.1.10
Destination Ip : 142.250.183.14
protocol : 6
Packet Size : 60 bytes
-----------------------
Source Ip : 8.8.8.8
Destination Ip : 192.168.1.10
protocol : 1
Packet Size : 84 bytes
-----------------------
```

Result:

Thus, the ping program and packet Sniffing program were successfully run using raw sockets.